

CONTACT	Georgia Institute of Technology Sacramento, California, United States	<i>E-mail:</i> ethan.villalovoz@gatech.edu <i>Links:</i> Website , LinkedIn , Google Scholar
RESEARCH INTERESTS	Embodied intelligence, robot learning, and human-robot interaction.	
EDUCATION	Georgia Institute of Technology <i>M.S. in Computer Science</i> , Computational Perception and Robotics	Jan 2026 – May 2028 GPA: 4.0/4.0
	Washington State University <i>B.S. in Computer Science</i> , Minor in Mathematics	Aug 2021 – May 2025 GPA: 3.94/4.0
PRE-PRINTS	[P1] An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models. S. Tomar, A. Deshwal, E. Villalovoz , M. Fazzini, H. Cai, J.R. Doppa. <i>arXiv:2512.15076</i> , 2025.	
CONFERENCE PUBLICATIONS	[C1] Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach. A. Bacula, E. Villalovoz , D. Flynn, A. Mehta, H. Knight. <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> , 2023.	
EXPERIENCE	Georgia Institute of Technology, Atlanta, Georgia, United States Feb 2026 – Present <i>Graduate Student Researcher</i>, Advised by Animesh Garg Developing a video-to-action robot policy learning framework coupling WAN 2.2 video diffusion with action flow matching; building PyTorch/Hydra pipelines for LIBERO rollouts and ablations. Microsoft, Redmond, Washington, United States May 2026 – Jul 2026 <i>Software Engineer Intern</i>, Advised by Karunakar Vecham Built and evaluated an enterprise AI agent proof of concept in C#/.NET 10 with Azure AI Foundry and Azure OpenAI; closed or validated 22 of 24 assigned security findings in Test/UAT. Washington State University, Pullman, Washington, United States Jan 2024 – May 2025 <i>Undergraduate Research Assistant</i>, Advised by Janardhan Rao (Jana) Doppa Implemented evaluation code and reproducible benchmarks for BODE-GEN, a Bayesian prompt-optimization method for test-driven LLM code generation across HumanEval+ tasks. Carnegie Mellon University, Pittsburgh, Pennsylvania, United States Jun 2024 – Aug 2024 <i>Robotics Institute Summer Scholar</i>, Advised by Henny Admoni, Reid Simmons Developed a Bayesian reward-learning system for simulated household manipulation from state-correction and feature-clarification interactions. Google, Sunnyvale, California, United States May 2023 – Aug 2023 <i>Software Engineering Intern (STEP)</i>, Advised by Arun Tej Chennadi, Paul Valdez Developed 5 C++/SQL analytics jobs for internal database queue metrics and dashboard tools supporting engineering data workflows. Oregon State University, Corvallis, Oregon, United States Jun 2022 – Aug 2022 <i>NSF REU Fellow</i>, Advised by Heather Knight Designed geometric motion primitives and Python/Tkinter tools for expressive formations in human-robot interaction research.	

AWARDS & HONORS	CS Research Mentorship Program Scholar, Google Research	2023
	Selected for a structured mentorship program supporting undergraduate pathways into computer science research.	
	Generation Google Scholarship	2023
	Awarded for academic performance, leadership, and commitment to diversity, equity, and inclusion in computing.	
	NIH MARC Scholar - National Institutes of Health (T34)	2023 – 2025
	Selected for a two-year NIH-funded undergraduate research training program supporting scientific research, leadership development, and graduate-school preparation.	
TEACHING	CS 8803 O27: Computer Graphics in the AI Era	Fall 2026
	<i>Graduate Assistant, Georgia Institute of Technology</i>	
	CPT S 315: Introduction to Data Mining	Spring 2025
	<i>Undergraduate Teaching Assistant, Washington State University</i>	
	CPT S 350: Design and Analysis of Algorithms	Fall 2024
	<i>Undergraduate Teaching Assistant, Washington State University</i>	
	CPT S 355: Programming Language Design	Fall 2023
	<i>Undergraduate Teaching Assistant, Washington State University</i>	
	CPT S 121: Program Design and Development C/C++	Fall 2022
	<i>Undergraduate Teaching Assistant, Washington State University</i>	
OUTREACH	WSU MARC & MIRA Program (Invited Talk)	2025
	Invited to speak with undergraduate researchers about graduate-school applications and strategies for pursuing research opportunities.	
	WSU VCEA (College Ambassador)	2022 – 2024
	Represented Voiland College to industry partners, alumni, and prospective students.	
	CMU RISS RoboLaunch (Website Coordinator)	2024
	Updated the RoboLaunch website for a robotics outreach initiative featuring talks and interactive workshops.	